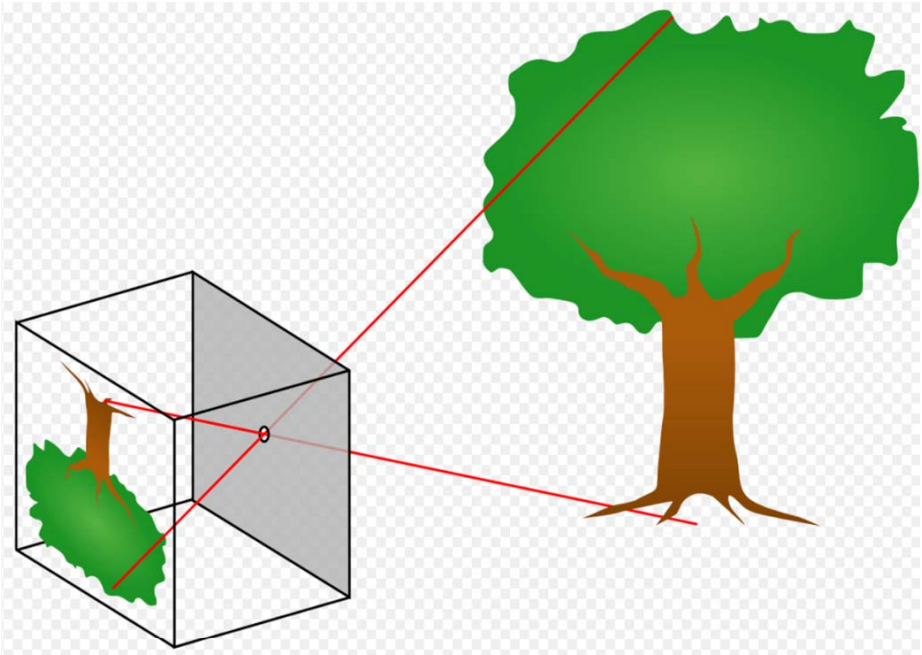
Computer Vision
Lecture 3: Digital Image

Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

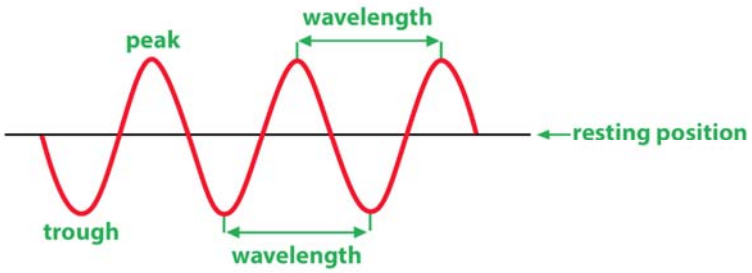
What is image?

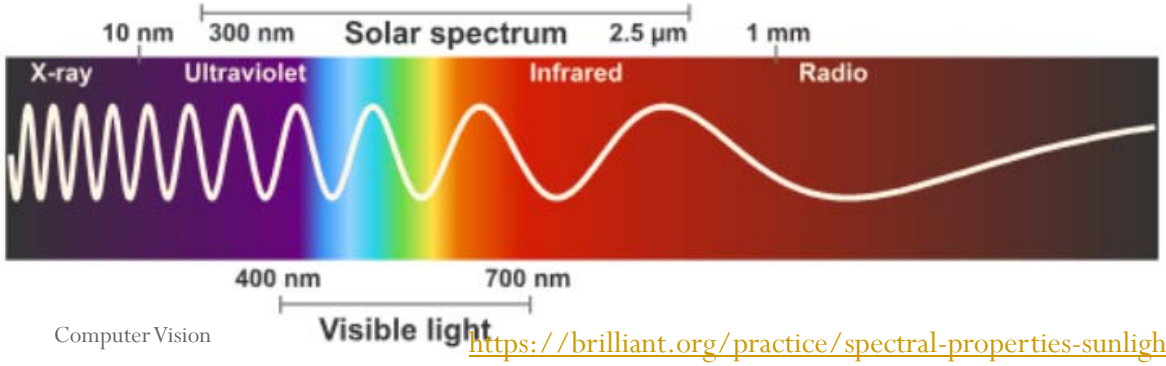
projection of a 3D scene into a **2D *projection plane***



Wavelength (λ)

- **Spatial period** of a periodic wave - the distance over which the wave's shape repeats, Measured in meters (m)

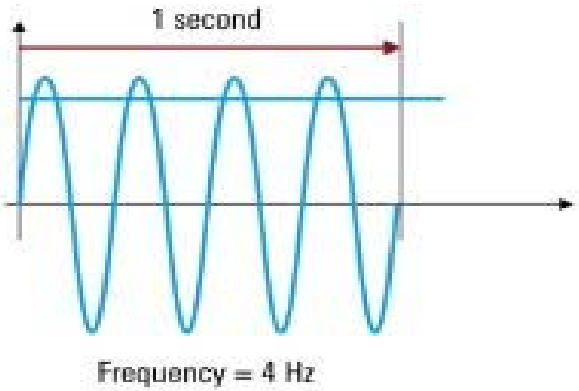




https://brilliant.org/practice/spectral-properties-sunlight/

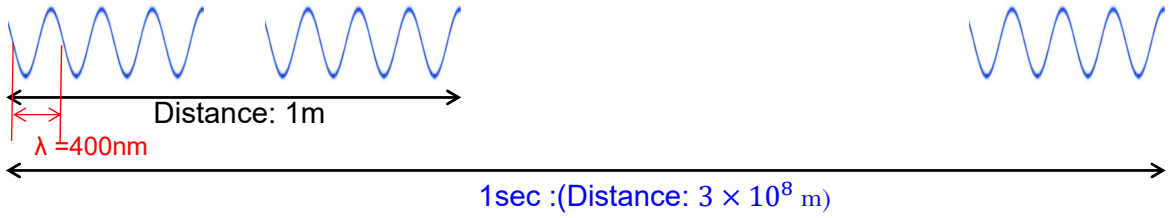
Frequency (f)

- The number of waves that pass a fixed point in one second.
- Measured in Hertz (Hz), where 1 Hz is one wave per second



Wavelength (λ) vs. Frequency (f)

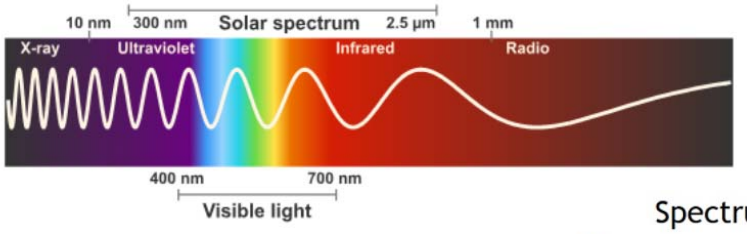
- **Ex**: λ =400nm
  - Speed of light: $c = 299{,}792{,}458\ m/s \approx 3 \times 10^8\ m/s$
  - $\frac{1}{\lambda} = \frac{1}{400\times10^{-9}} = \frac{1}{400} \times 10^9 = 2{,}500{,}000\ \ periods/m$
  - Frequency: $f = \frac{c}{\lambda} = \frac{3\times10^8}{400\times10^{-9}} = 750 \times 10^{12} = 750\ THz$

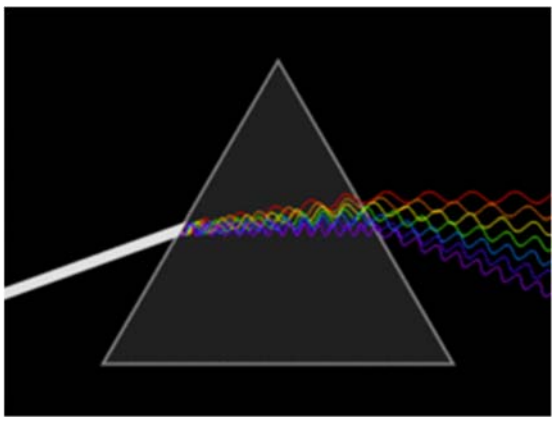


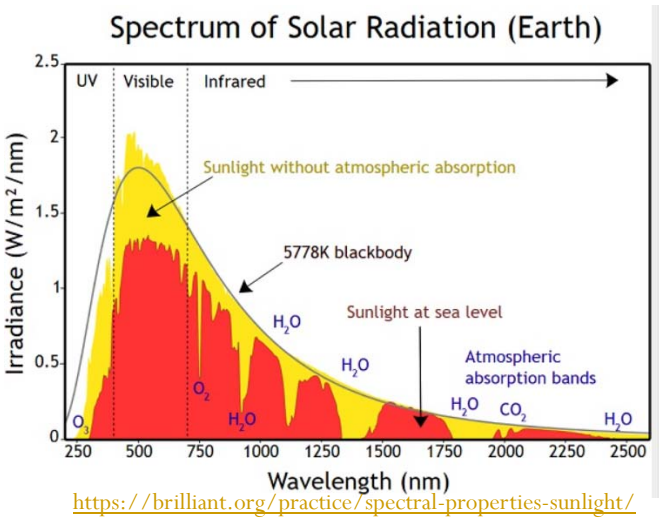
https://brilliant.org/practice/spectral-properties-sunlight/

Wavelength

- **Sun light**: About 99% of solar power reaching Earth's surface has a wavelength between 300 nm and 2.5 μm



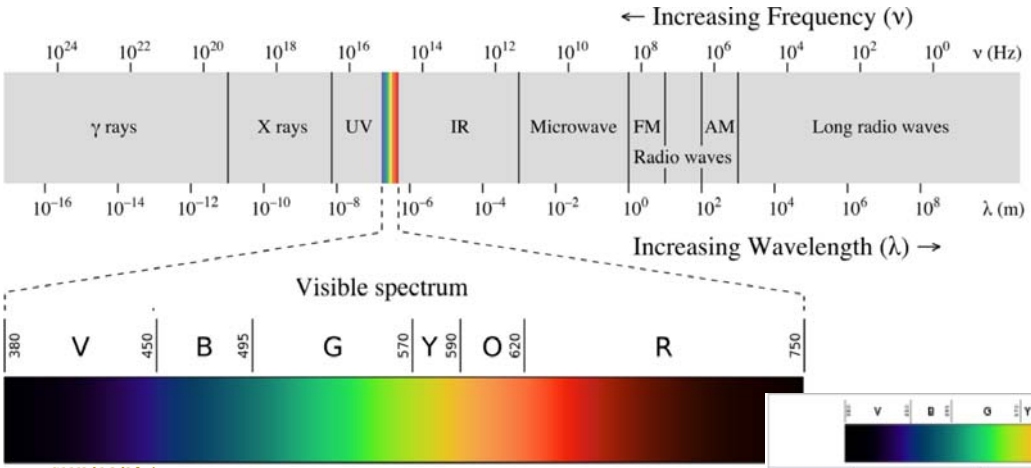




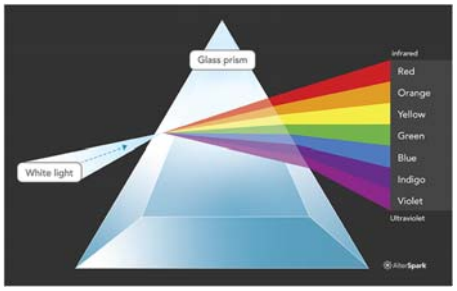
https://brilliant.org/practice/spectral-properties-sunlight/

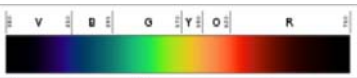
Visual light

- Human eye respond to wavelengths from about λ=380 to about 750 nm



sunlight/

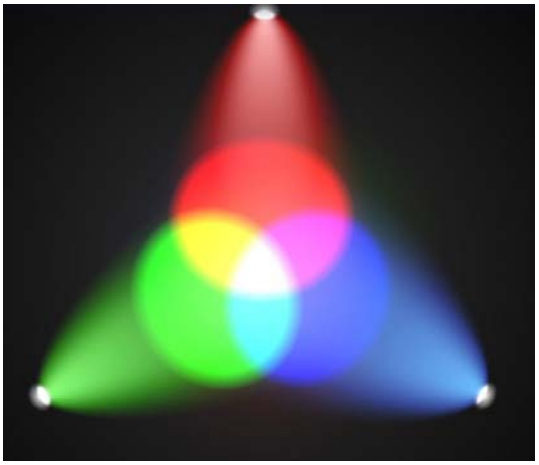


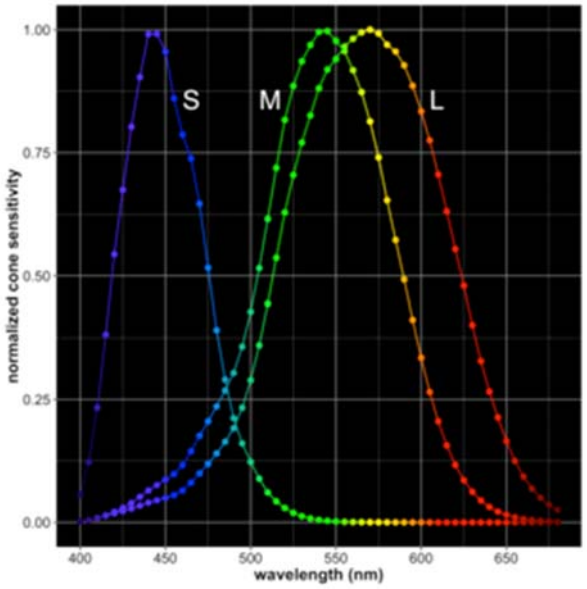


Color	Wavelength	Frequency	Photon energy
Violet	380–450 nm	670–790 THz	2.75–3.26 eV
Blue	450–485 nm	620–670 THz	2.56–2.75 eV
Cyan	485–500 nm	600–620 THz	2.48–2.56 eV
Green	500–565 nm	530–600 THz	2.19–2.48 eV
Yellow	565–590 nm	510–530 THz	2.10–2.19 eV
Orange	590–625 nm	480–510 THz	1.98–2.10 eV
Red	625–700 nm	400–480 THz	1.65–1.98 eV

R/G/B light

- Primary color of light





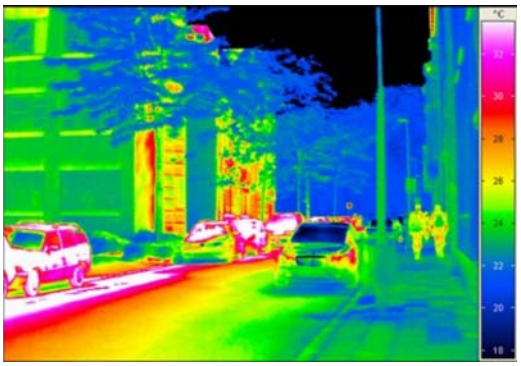
Normalized cone spectral sensitivity

https://en.wikipedia.org/wiki/Primary_color#Color-space_primaries

IR(InfraRed) Image



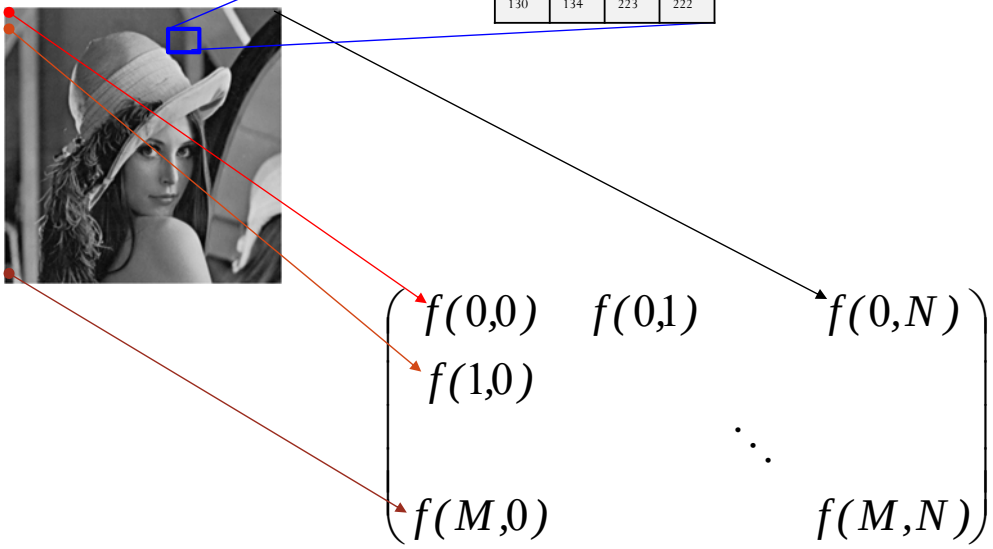






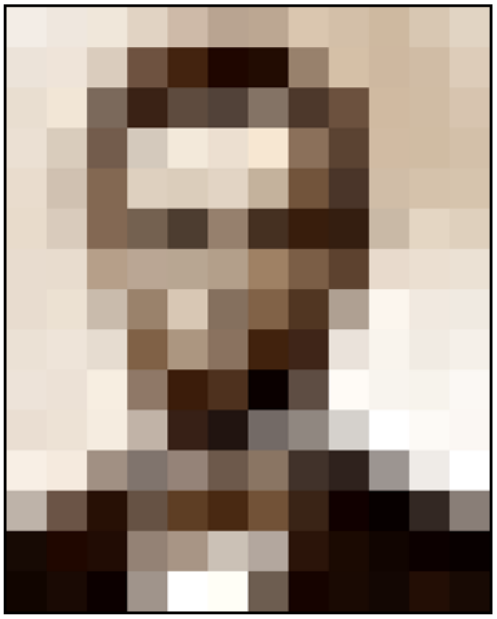
Gray-scale (Black & white) Image

- Represented by 8 bits /pixel (unsigned character) : 0~255



Gray-scale (Black & white) Image

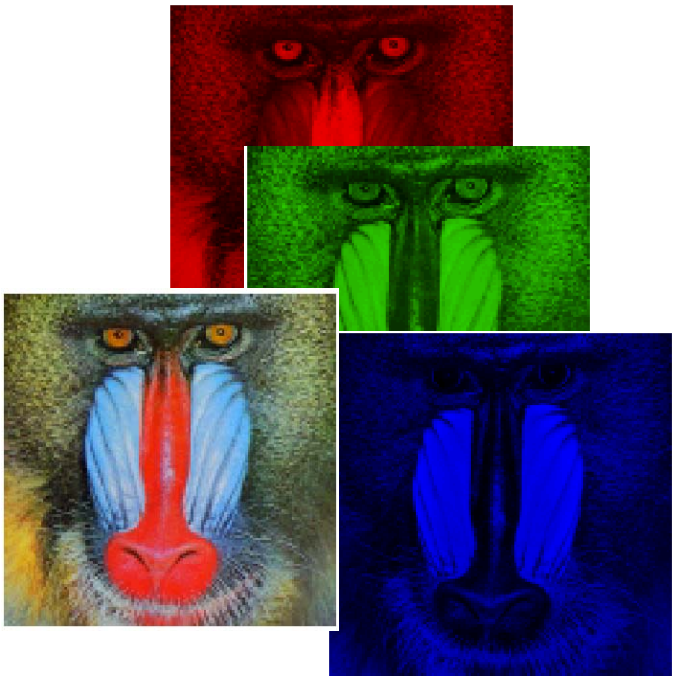
- Represented by 8 bits /pixel (unsigned character) : 0~255

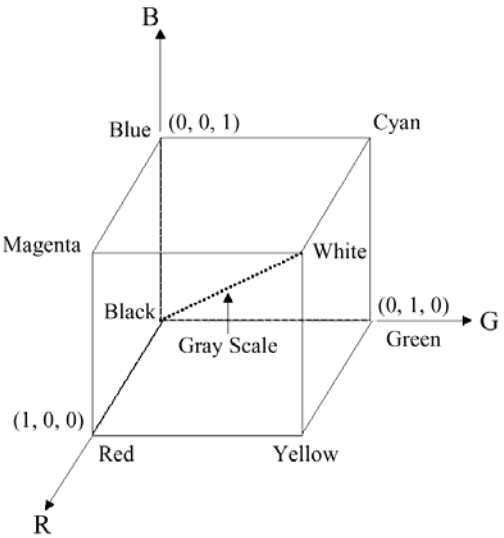


243	239	240	225	206	185	188	218	211	206	216	225
242	239	218	110	67	31	34	152	213	206	208	221
243	242	123	58	94	82	132	77	108	208	208	215
235	217	115	212	243	236	247	139	91	209	208	211
233	208	131	222	219	226	196	114	74	208	213	214
232	217	131	116	77	150	69	56	52	201	228	223
232	232	182	186	184	179	159	123	93	232	235	235
232	236	201	154	216	133	129	81	175	252	241	240
235	238	230	128	172	138	65	63	234	249	241	245
237	236	247	143	59	78	[illegible]	94	255	248	247	251
234	237	245	193	55	33	115	144	213	255	253	251
248	245	161	128	149	109	138	65	47	156	239	255
190	107	39	102	94	73	114	58	[illegible]	[illegible]	51	137
[illegible]	[illegible]	[illegible]	148	168	203	179	43	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	160	255	255	109	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Color Images

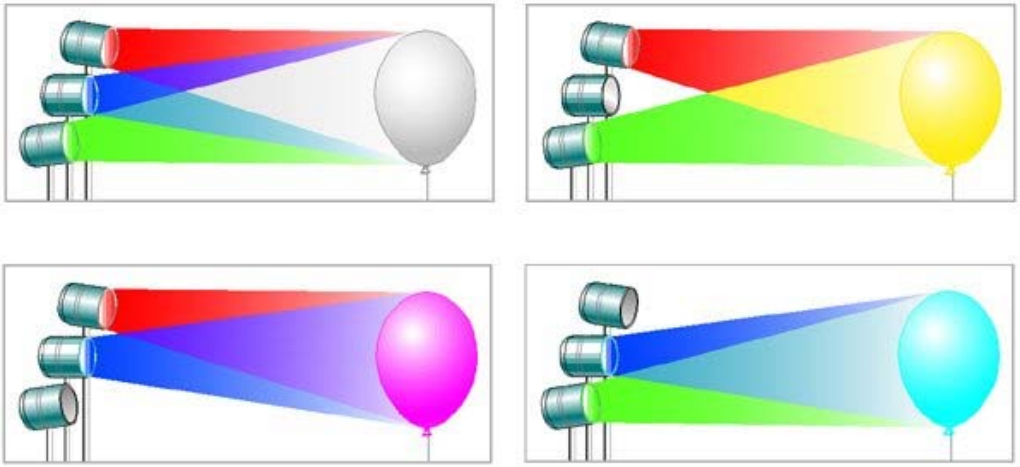
- RGB representation : $f_R(x,y)$ $f_G(x,y)$ $f_B(x,y)$
- Represented by 3x8=24 bits /pixel , 8bits/color : 0~255





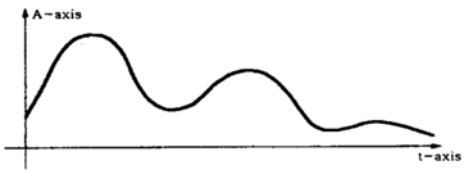
Color Images

	Nominal Range	White	Yellow	Cyan	Green	Magenta	Red	Blue	Black
R	0 to 255	255	255	0	0	255	255	0	0
G	0 to 255	255	255	255	255	0	0	0	0
B	0 to 255	255	0	255	0	255	0	255	0



Digital Signal

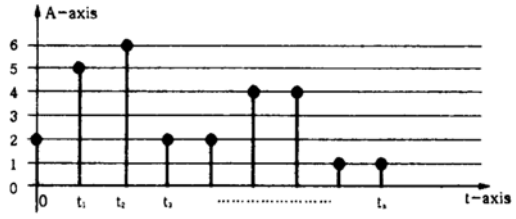
- Analog-to-Digital Converter
  - Sampling(표본화)
  - Quantization(양자화)



연속 신호



이산 신호



디지털 신호

{0,1,2,3,4,5,6}

Binary Coding



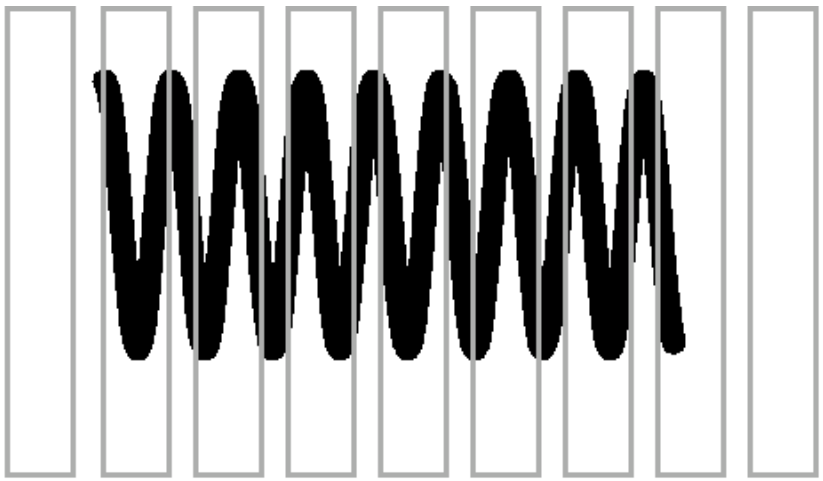
이진 디지털 신호

{0,1}

Sampling



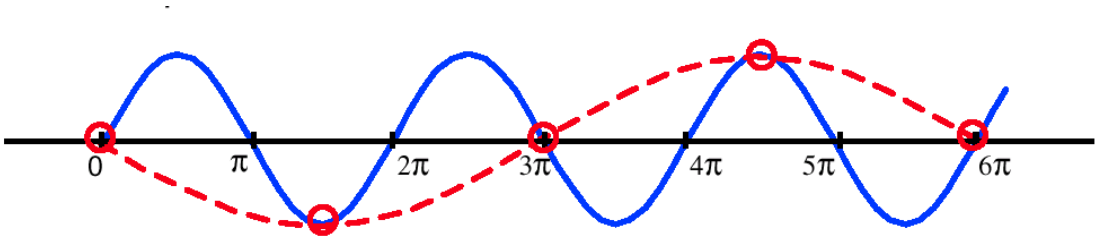
Sampling



Sampling Theorem

- 표본화 하고자 하는 연속신호의 최대주파수를 f_{max}(Hz)라 할 때, 표본화 율(sampling rate)를 $2xf_{max}$ 이상으로 하면, 그 결과로 형성되는 디지털 신호로 원래의 연속신호를 왜곡 없이 완벽하게 복원할 수 있다.

$$f_{sampling} \geq 2f_{max}$$

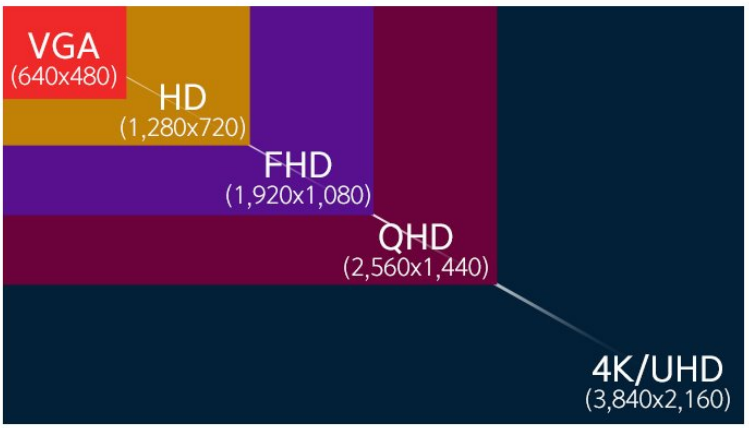


Sampling : resolution



Resolution

- VGA : 640 x 480
- HD : 1280 x 720 : 720p
- FHD : 1920 x 1080 : 1080p
- UHD : 4K : 3840 x 2160
- 8K : 7680 x 4320
- 16K : 15360 x 8640



16K 인 경우,

- 15360 x 8640 =132,710,400 pixels
- 1 image: 3(channels)x8bitsx132,710,400 pixels
  = 3,185,049,600bits= 398,131,200 bytes = 0.39 GBytes
- 1sec Video: 30(frames)x3(channels)x8bitsx132,710,400 pixels
  = 95,551,488,000 bits = 11,943,936,000 bytes = 11.9 GBytes

Quantization: the number of bits


1 bit


2 bit (0~3)

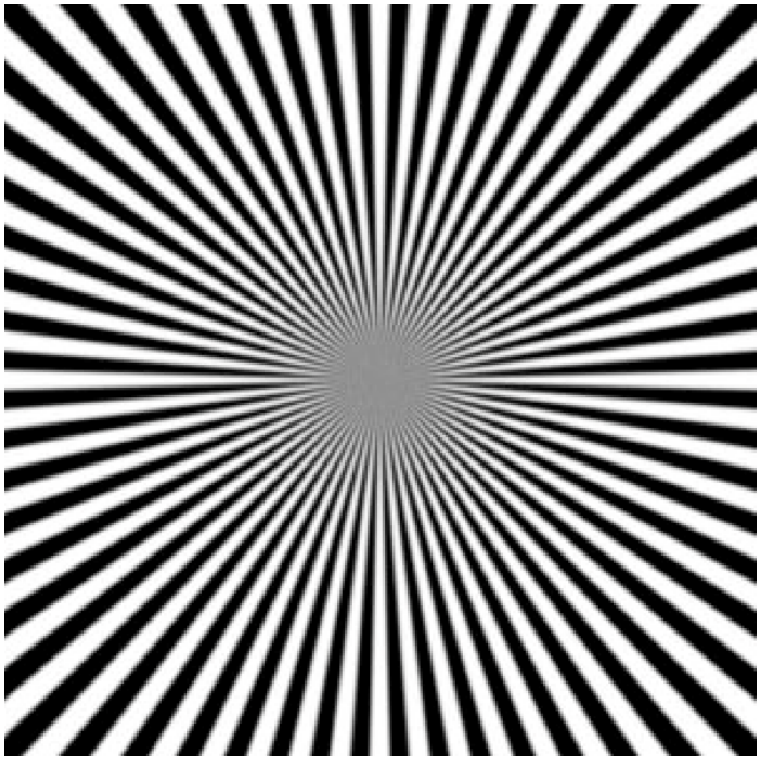

3 bit (0~7)


4 bit (0~15)

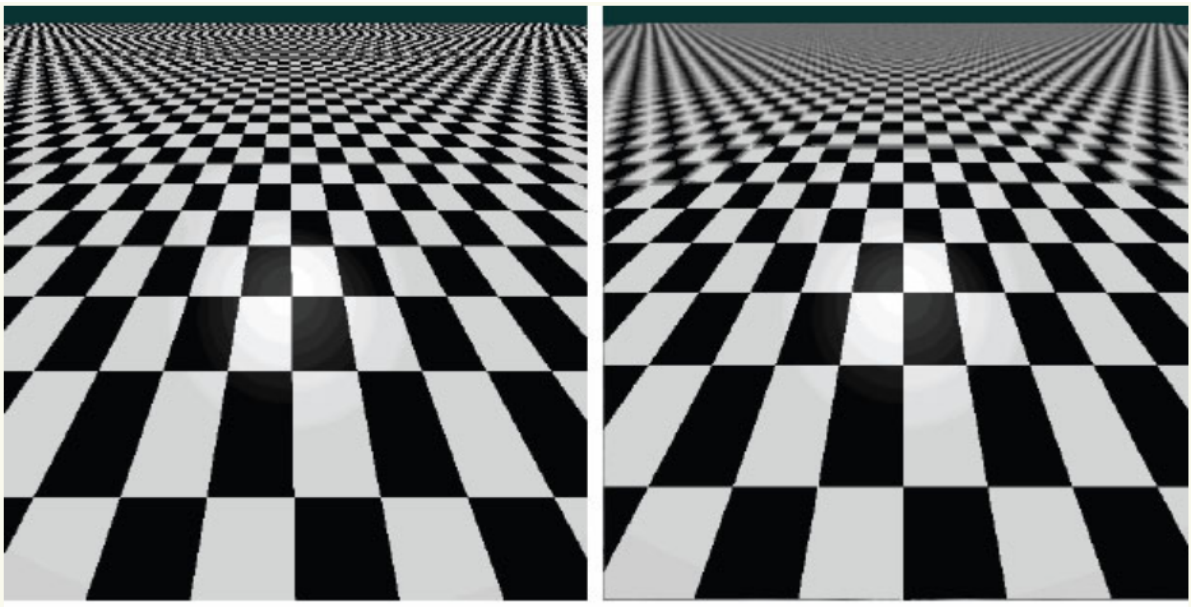

6 bit (0~63)


8 bit (0~255)

Spatial Aliasing

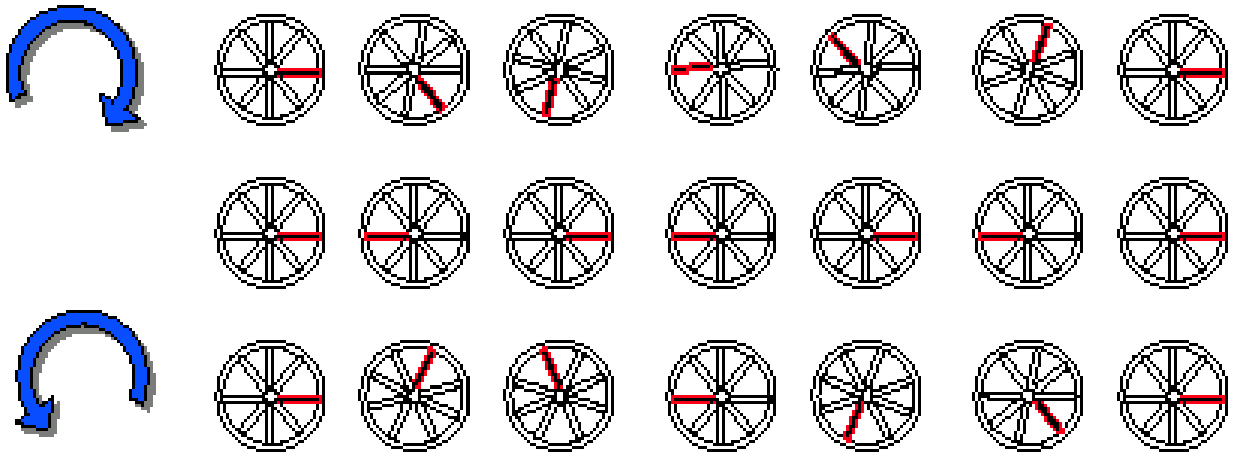


Spatial Aliasing



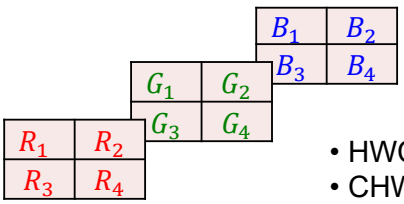
http://mielliott.github.io/index.html

Temporal Aliasing



Color Spaces & Channels

- Dimension Layouts: Grayscale vs. Color Images
  - Grayscale : 2D Tensor → $H \times W \times 1$: Only contains intensity
  - Color (RGB): 3D Tensor → $H \times W \times 3$: Combination of Red, Green, and Blue matrices.
- The Critical Trap: RGB vs. BGR
  - OpenCV (cv2): Loads images in BGR order.
  - PyTorch / PIL / Matplotlib: Use the standard RGB order.
- Memory Alignment (Layouts)
  - HWC: (Height, Width, Channel) → Default for OpenCV & TensorFlow.
  - CHW: (Channel, Height, Width) → Required for PyTorch tensor operations.



• HWC: [(R1, G1, B1), (R2, G2, B2), (R3, G3, B3), (R4, G4, B4)]
• CHW: [R1, R2, R3, R4, G1, G2, G3, G4, B1, B2, B3, B4]

Image Formats & Compression

- PNG (Lossless Compression)
  - Supports 4 channels (RGBA) including Transparency (Alpha).
  - Best for: Ground-truth labels, segmentation masks, and medical imaging.
- The JPEG / JPG (Lossy Compression)
  - Discards high-frequency spatial frequencies that are less visible to human eyes.
  - Drastically reduces file size but introduces compression artifacts (pixel value distortion).
  - Warning: Avoid using JPEG for tasks that require pixel-level precision.
- EXIF(Exchangeable Image File Format, 교환 가능한 이미지 파일 형식) Metadata & Orientation
  - Stores camera model, exposure settings, and Orientation Flags.
  - Neglecting EXIF orientation might cause unexpected image rotation when parsing raw pixels.

Thank You
Next : Color Transform

